# TERM-DEPOSIT CAMPAIGN TARGETING

## Predictive Analytics Portfolio Case Study

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|---|---|
| **BUSINESS CHALLENGE** | Prioritise prospects for a costly outbound telemarketing campaign. |
| **ANALYTICAL APPROACH** | Descriptive analysis, decision-tree exploration and supervised classification. |
| **TECHNOLOGY** | Python, pandas, scikit-learn and matplotlib. |

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**PORTFOLIO NOTE**

This report is a professionally revised portfolio adaptation of an individual academic project. The raw dataset is not distributed. All results are based on the supplied historical sample and should be validated before operational use.

# Executive summary

A banking business wants to reduce the cost of unproductive telemarketing calls while continuing to reach customers who may subscribe to a fixed-term deposit product. The analysis uses 4,000 historical call records, of which 800 (20%) resulted in a subscription.

A leakage-sensitive modelling design was used: call duration was analysed descriptively but excluded from prediction because it is unavailable before a future call. Three classifiers were compared using five-fold stratified cross-validation on the training sample, followed by one evaluation on a 30% holdout set.

| 4,000 | 20.0% | 0.534 | 0.764 |
|---|---|---|---|
| historical calls | subscription rate | RF cross-validated F1 | RF holdout ROC-AUC |

## Decision

The Random Forest provided the strongest overall discrimination and the best balance between precision and recall. On the holdout sample it achieved 50.4% precision, 54.2% recall, 52.2% F1 and 76.4% ROC-AUC. Accuracy alone was not used for selection because an 80% majority-class benchmark identifies no subscribers.

## Commercial interpretation

The model is most useful as a prioritisation tool rather than a binary rejection system. On the holdout sample, the top-scored 10% of prospects converted at 64.2% and contained 32.1% of all subscribers. These figures illustrate the value of probability ranking, but they are historical estimates rather than guaranteed future performance.

## Recommended operating model

- Rank prospects by predicted subscription probability and align the calling threshold with available call-centre capacity.
- Monitor conversion, false-positive volume, subgroup performance and probability calibration after each campaign.
- Retrain only when sufficient new outcomes are available and performance deterioration is evidenced.

**KEY CAUTION**

Observed relationships are predictive associations, not causal effects. For example, high conversion among some age or job groups should not be interpreted as proof that those characteristics cause subscription.

# 1. Business objective and analytical design

## 1.1 Business question

Which prospects should be prioritised for contact when call-centre capacity is limited and unproductive calls create avoidable staffing cost? The required output is a probability score that supports ranked targeting, together with transparent evidence on performance and limitations.

## 1.2 Data used

The dataset contains 4,000 historical call records. Fourteen pre-call predictors were used: eight categorical variables (job, marital status, education, default, housing loan, personal loan, contact channel and previous campaign outcome) and six numeric variables (age, balance, day, campaign contacts, days since prior contact and previous contacts).

Call duration was excluded from all predictive models. Although it is strongly associated with the outcome, it becomes known only after a call has occurred and would therefore introduce target leakage into a prospecting model.

# 2. Descriptive findings

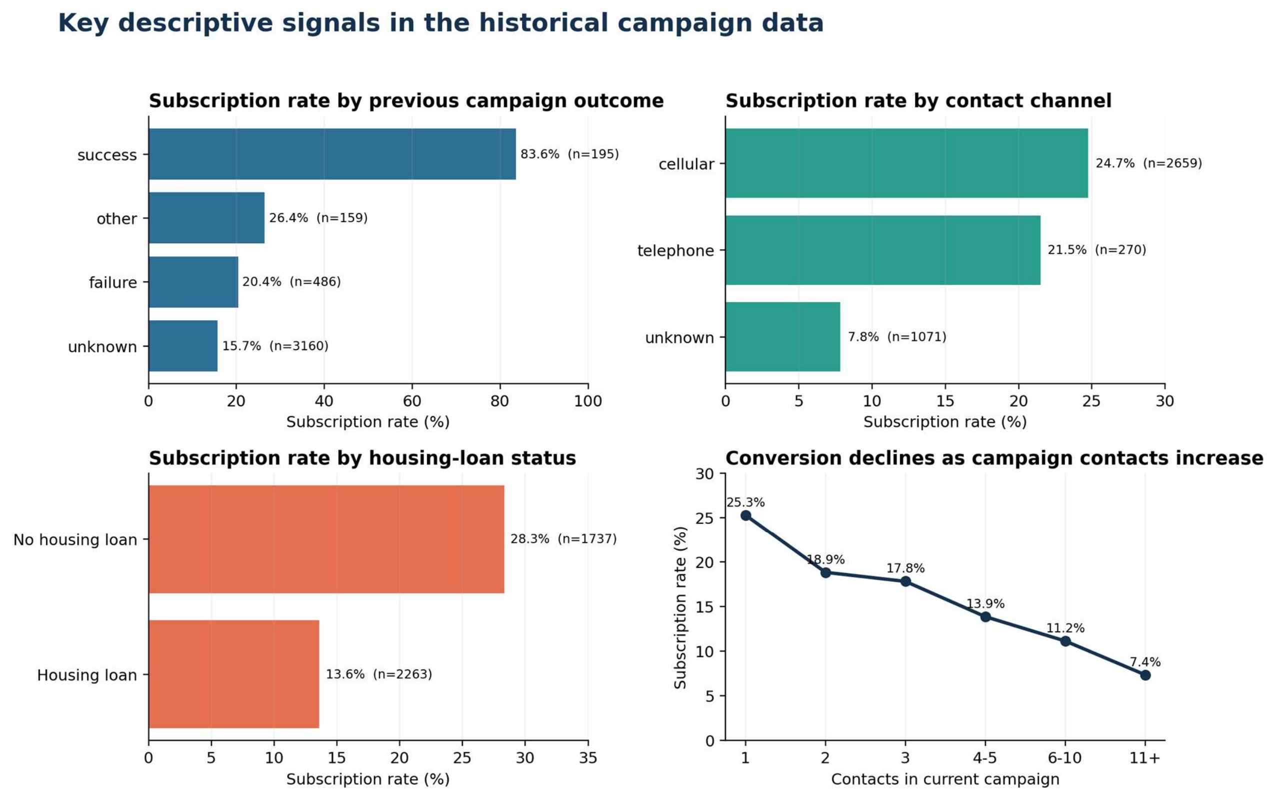



*Figure 1. Conversion rates are descriptive and should be interpreted alongside group size.*

The clearest historical signal was a successful previous campaign outcome: 83.6% of this group subscribed, compared with 15.7% when previous outcome was unknown. Known contact channels also performed materially better than an unknown channel. Customers without a housing loan converted at 28.3%, compared with 13.6% for those with one.

Conversion declined as the number of campaign contacts increased, from 25.3% at the first contact to 7.4% after 11 or more contacts. This supports tighter contact-frequency controls, but does not by itself establish that additional calls cause lower conversion because harder-to-convert prospects may naturally receive more attempts.

# 3. Modelling and evaluation

## 3.1 Evaluation design

A stratified 70/30 split produced a 2,800-record development sample and a 1,200-record untouched holdout sample. Model selection used five-fold stratified cross-validation within the development sample. Categorical variables were one-hot encoded; numeric variables were standardised for Logistic Regression and passed through unchanged for the tree-based models.

| Model | Configuration | Purpose in comparison |
|---|---|---|
| **Decision Tree** | Entropy; depth 4; minimum leaf 7 | Interpretable, high-precision baseline |
| **Logistic Regression** | Scaled numeric inputs; balanced class weights | Linear probability model with higher recall |
| **Random Forest** | 200 trees; minimum leaf 5; balanced class weights | Non-linear ensemble for balanced discrimination |

## 3.2 Cross-validation results

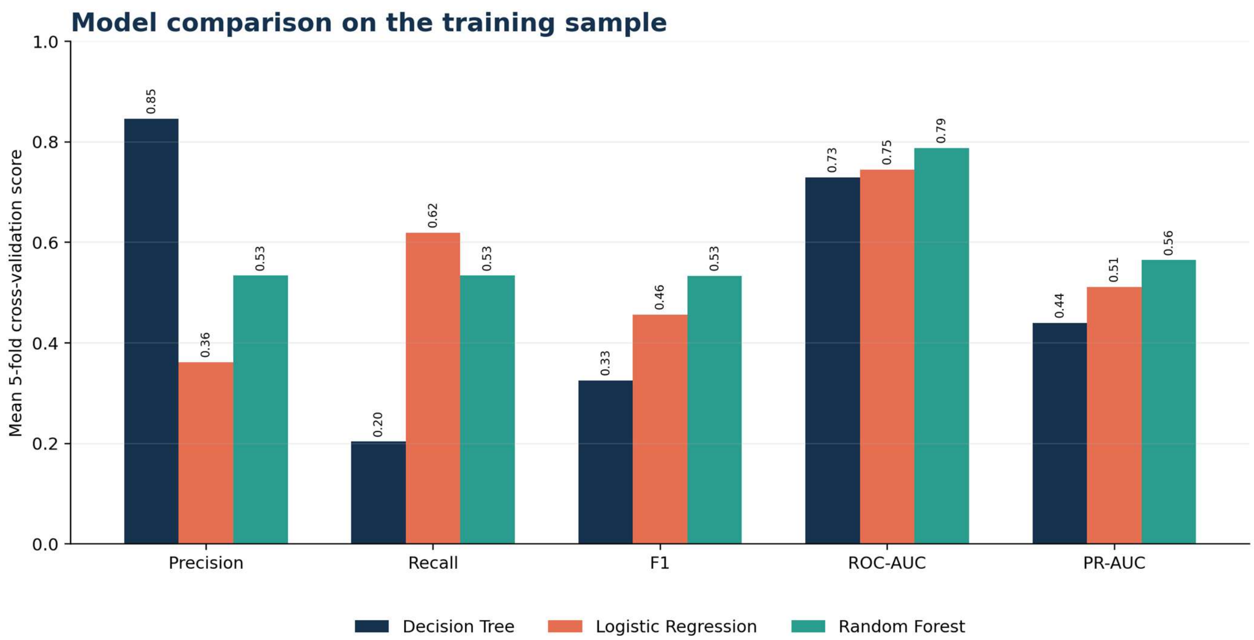


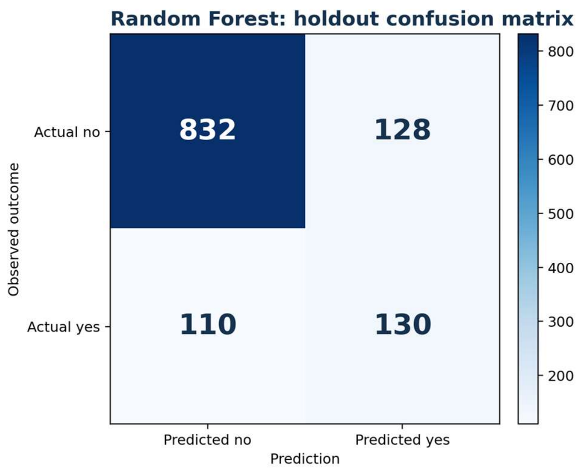

*Figure 2. Mean five-fold scores on the development sample. The positive class is subscription = yes.*

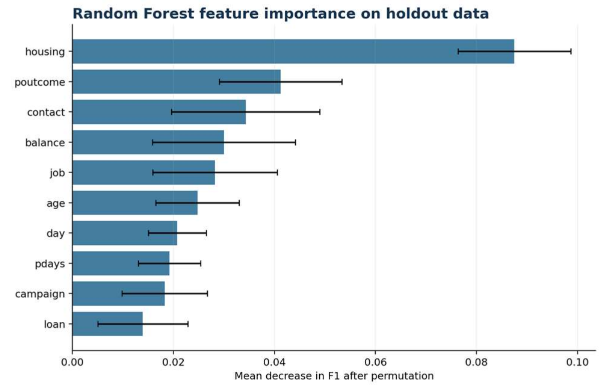
| Model | Accuracy | Precision | Recall | F1 | ROC-AUC | PR-AUC |
|---|---|---|---|---|---|---|
| **Majority benchmark** | 0.800 | - | 0.000 | 0.000 | 0.500 | 0.200 |
| **Decision Tree** | 0.833 | 0.847 | 0.204 | 0.325 | 0.729 | 0.440 |
| **Logistic Regression** | 0.705 | 0.362 | 0.620 | 0.457 | 0.745 | 0.511 |
| **Random Forest** | **0.814** | **0.534** | **0.534** | **0.534** | **0.788** | **0.565** |

# 4. Final model assessment

## 4.1 Holdout performance

After model selection, the Random Forest was trained on the full development sample and evaluated once on the holdout data. It correctly identified 130 subscribers, missed 110, and generated 128 false-positive calls. This produced 50.4% precision, 54.2% recall and 52.2% F1.




*Figures 3-4. Holdout confusion matrix and grouped permutation importance measured by decrease in F1.*

## 4.2 Interpretation

The model improved substantially on the majority benchmark for identifying subscribers, although holdout accuracy (80.2%) was only marginally above the benchmark (80.0%). This is why class-specific metrics and the confusion matrix are more informative than accuracy for this imbalanced problem.

Permutation importance indicates that housing-loan status, previous campaign outcome, contact channel, balance, job and age contributed most to holdout F1. These values reflect model reliance within this dataset; they do not measure causal impact and may vary as customer behaviour changes.

**MODEL GOVERNANCE**

The score should guide contact priority, not automatically exclude customers from access to a product. Before deployment, the organisation should review subgroup error rates, legal constraints and the commercial cost of false positives and false negatives.

# 5. From model score to campaign action

## 5.1 Capacity-based targeting

A probability ranking is more flexible than a fixed yes/no prediction. The call centre can select the highest-scored prospects until its available capacity is reached. On the holdout sample, contacting the top-ranked 10% captured 32.1% of all subscribers at a 64.2% conversion rate; contacting the top 30% captured 62.1% at a 41.4% conversion rate.

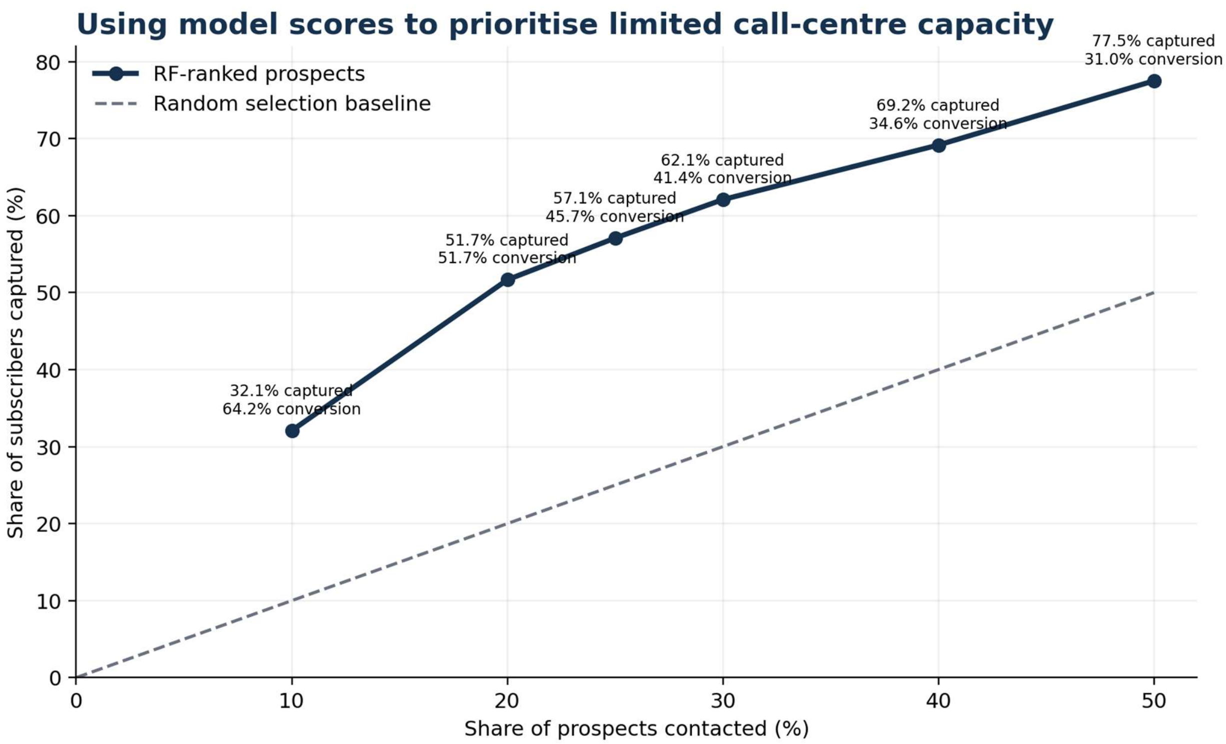



*Figure 5. Historical holdout ranking performance. Future campaign results may differ.*

## 5.2 Deployment workflow


*Figure 6. The model produces a probability score that is converted into a ranked campaign list.*

## 5.3 Technical implementation

- Input: a CSV containing the same 14 pre-call fields used during training; the target and call duration are not required.
- Output: predicted class and predicted subscription probability for each prospect.
- Operational control: select the call threshold or top-ranked share based on staff capacity and expected campaign economics.
- Monitoring: log predictions, outcomes, contact costs, conversion and subgroup performance for every campaign cycle.

# 6. Recommendations, limitations and conclusion

## 6.1 Recommendations

**Prioritise probability-ranked lists:** Use predicted probability to sequence calls, rather than relying on broad demographic rules or a single fixed classification threshold.

**Protect call-centre efficiency:** Apply a contact-frequency policy and review prospects with unknown contact information, which historically showed low conversion.

**Use previous success carefully:** Prior campaign success is a strong signal, but the segment is small; preserve a broader prospect pool to avoid over-concentration.

**Measure campaign economics:** Estimate the cost of a call, expected margin from a subscription and the cost of a missed customer. Use these values to select the operating threshold.

**Monitor and refresh:** Track drift, calibration, precision, recall and subgroup performance. Retrain only when new labelled outcomes provide evidence for an update.

## 6.2 Limitations

- The dataset contains call records rather than unique customer identifiers, so repeated contacts cannot be modelled at customer level.
- The sample reflects historical campaign conditions and may not represent future economic or behavioural conditions.
- No explicit call-cost, product-margin or customer-lifetime-value data were available, preventing direct profit optimisation.
- Some categories are labelled unknown, which may represent operational data-quality issues rather than genuine customer characteristics.
- Results are internally validated only; an external or later-period validation sample would strengthen deployment confidence.

## 6.3 Conclusion

This project demonstrates an end-to-end business analytics workflow: defining a commercial problem, preventing leakage, translating descriptive findings into model features, comparing classifiers with metrics suited to class imbalance, and converting predictions into an operational ranking strategy.

The Random Forest is the preferred model because it achieved the strongest cross-validated F1 and ranking performance while maintaining a more practical precision-recall balance than the alternatives. Its value lies in focusing scarce calling capacity on higher-probability prospects, supported by ongoing monitoring and disciplined threshold management.

**PORTFOLIO TAKEAWAY**

The project combines Python modelling with business interpretation: the final recommendation is not simply “use Random Forest”, but to embed calibrated probability scores within a monitored, capacity-aware campaign process.